

IBRA, Oman

WMO: 412650

Lat: 22.8094N

Lon: 58.4606E

Elev: 469

StdP: 95.82

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.1	(c) 13.4	(d) -7.0	(e) 2.2	(f) 26.7	(g) -4.6	(h) 2.7	(i) 27.2	(j) 7.9	(k) 22.7	(l) 6.3	(m) 23.3	(n) 1.5	(o) 320	(p) 0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 13.2	(c) 44.3	(d) 21.3	(e) 43.3	(f) 21.5	(g) 42.3	(h) 21.6	(i) 26.6	(j) 34.2	(k) 26.1	(l) 34.1	(m) 25.7	(n) 33.8	(o) 4.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.5	29.3	24.3	20.4	29.4	23.7	19.6	29.5	86.7	33.3	84.4	34.1	82.3	33.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	6.7	5.9	9.1	46.4	1.2	0.8	8.2	46.9	7.5	47.4	6.8	47.8	6.0	48.4
			3.7	27.0	1.5	0.7	2.6	27.5	1.8	27.9	0.9	28.3	-0.2	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	19.9	21.8	25.6	30.0	33.8	35.4	35.1	34.0	32.3	29.3	24.6	21.3	
	DBStd	5.87	1.81	2.45	2.35	2.22	1.86	2.10	2.44	2.39	1.88	1.75	1.84	1.92	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	12	7	3	0	0	0	0	0	0	0	0	0	2	
	CDD10.0	6794	306	330	484	601	737	762	777	744	668	597	439	349	
	CDD18.3	3764	54	99	226	351	478	512	519	486	418	338	189	93	
	CDH23.3	53871	411	966	2602	4906	7657	8462	8259	7393	6079	4484	1902	749	
CDH26.7	34263	54	307	1301	2982	5232	6079	5834	5031	3889	2615	782	158		
Wind	WSAvg	2.4	2.0	2.5	2.6	2.8	2.9	2.9	2.6	2.5	2.4	2.2	2.0	1.9	
Precipitation	PrecAvg	133	6	18	20	25	5	11	10	10	5	6	9	8	
	PrecMax	231	53	118	119	98	25	99	49	50	46	60	96	67	
	PrecMin	13	0	1	0	2	0	0	0	0	0	0	0	0	
	PrecStd	69	12	30	30	29	7	23	14	11	9	14	22	19	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	33.2	37.0	40.3	44.8	45.7	45.6	44.2	42.5	39.3	34.6	31.3	
		MCWB	15.3	16.7	17.5	19.1	20.9	20.6	22.5	22.1	20.9	18.7	18.4	16.4	
	2%	DB	28.1	31.7	35.6	39.0	42.6	44.4	44.1	43.0	41.1	38.1	33.1	29.6	
		MCWB	16.1	16.7	17.4	19.0	20.6	20.6	22.2	22.3	20.9	18.8	17.7	15.9	
	5%	DB	26.8	30.1	34.2	37.9	41.2	43.4	42.9	42.0	39.9	37.0	32.0	28.4	
		MCWB	16.0	16.6	17.3	18.7	20.5	20.7	22.4	22.3	21.2	18.6	17.3	16.2	
	10%	DB	25.6	28.5	32.8	36.7	40.1	42.2	41.8	40.8	38.7	35.7	30.9	27.2	
		MCWB	15.7	16.4	17.1	18.5	20.5	20.8	22.7	22.9	21.2	18.2	17.3	16.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	21.2	22.4	24.3	26.2	27.5	27.2	26.8	26.1	23.9	22.7	20.6
			MCDB	24.5	25.2	26.7	28.6	31.7	32.6	36.5	35.1	34.2	29.9	29.0	24.2
2%		WB	18.9	19.7	21.5	23.0	25.5	26.7	26.5	26.2	25.1	23.1	21.1	19.7	
		MCDB	23.2	24.7	26.2	29.0	31.4	32.4	35.7	34.9	32.5	29.1	25.7	23.7	
5%		WB	18.1	19.0	20.5	22.2	24.9	26.0	26.0	25.8	24.5	22.4	20.3	19.1	
		MCDB	22.9	24.0	26.8	28.9	31.2	32.8	35.0	34.7	32.4	28.9	25.6	23.4	
10%		WB	17.3	18.2	19.5	21.4	24.2	25.4	25.5	25.2	23.9	21.6	19.6	18.3	
		MCDB	22.4	24.4	27.0	29.6	31.9	32.9	34.4	34.0	32.3	28.9	25.6	23.4	
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.7	13.3	12.9	12.9	13.2	13.0	13.1	13.2	13.0	12.0	11.8	
		MCDBR	13.4	14.6	14.9	13.8	13.9	14.4	13.8	13.8	13.8	14.0	13.1	13.4	
		MCWBR	4.5	4.8	4.7	3.8	4.0	4.2	3.3	2.9	3.6	4.0	4.6	4.5	
	5% WB	MCDBR	11.3	12.0	11.6	12.0	12.8	13.1	12.8	12.8	13.3	12.0	11.1	10.2	
		MCWBR	4.2	4.4	3.9	4.1	4.0	3.9	2.8	2.9	3.3	3.6	4.0	3.7	
Clear-Sky Solar Irradiance	taub	0.381	0.421	0.480	0.542	0.582	0.659	0.737	0.676	0.561	0.439	0.404	0.379		
	taud	2.293	2.143	1.955	1.835	1.753	1.625	1.528	1.621	1.814	2.097	2.232	2.295		
	Ebn at Noon	877	862	827	782	745	685	633	673	747	832	845	864		
	Edh at Noon	120	148	186	214	231	261	287	262	212	153	126	115		
All-Sky Solar Radiation	RadAvg	4.73	5.53	6.39	7.07	7.65	7.35	6.96	6.75	6.48	5.89	4.93	4.47		
	RadStd	0.19	0.21	0.32	0.30	0.13	0.29	0.33	0.19	0.12	0.14	0.14	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.23	N/A	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96

Nomenclature: See separate page